Tuesday Jan 21 (Command Lines Cont. and Git Overview)

Git: version control system, set of commands for managing a project

GitHub: web platform, account-based, use git commands to sync your projects with the web server, add collaborators, etc.

sudo apt install git	Tell package manager (apt) to install git package
git init	Initializes a local git repository
git status	Shows your working branch, any files to be added, changes to be committed, etc
git remote add **origin** <link to GitHub repository> ex. git remote add origin https://github.com/pedrampa/git practice.git	Syncs local repository to a remote GitHub repository. Whenever you see origin/... later, it refers to your remote repository origin/main: your remote branch main: your local branch
git clone <link to GitHub repository>	Create new local repository which is synced to remote GitHub repository
git checkout <branch-name>	Switch to another branch which currently exists in the repository
git checkout **-b** <branch-name>	Create a new branch, and switch to it
git pull	Pull changes from origin/branch into your local branch
git add <file-names>	Tells branch to keep track of changes to the selected files or git add . to add all untracked files
git commit -m “My commit message”	Attach a descriptive message relating to your changes to any file(s)
git push (1st time: git push -u origin <branch-name>)	Pushes your local changes to remote. If it’s a new branch, you need to set it to point to origin using git push -u origin
git merge <other-branch>	Merges changes from other-branch into the current branch

These are just a summary, there are many more commands

General Workflow:

1) git fetch origin
- pulls latest info from remote (origin), changes from your collaborators
2) git checkout (-b) <name of branch>

- typically on a project with multiple collaborators, you should work on a separate branch, then git merge into 'main' when your changes are ready & reviewed

3) git pull
4) Work on the branch, making your changes to the project
5) git push

Tasks:

- Practice Unix commands
- Refer to exercise from the prof's e-textbook, **section 5.5: GitHub and Version Control**
- Version control practice:
  - Create a GitHub account (if you don't have one)
  - Create a repository on GitHub
  - Clone that repository to your local device using git commands in the terminal
  - When you push for the first time, you will be asked for authentication
    - if using VS Code, the easiest way is to allow VS Code to open a browser where you can authenticate through GitHub website
    - Otherwise, use SSH token or PAT
  - Practice adding files, committing, pull/push, create/checkout different branches, etc.